

FOOT MUSCLES

PLAN

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Objectives:

- To understand the muscle insertions.
- To be able to classify these muscles according to their actions.
- To understand the innervation of the muscles.

I. INTRODUCTION:

The muscles of the foot are divided into two regions:

- A dorsal region covered by the dorsal fascia of the foot.
- A plantar region: which is divided into three compartments: lateral, medial, and intermediate

II. DESCRIPTIVE ANATOMY:

A. MUSCLES OF THE DORSAL REGION: There are two:

- Extensor digitorum brevis.
- Extensor hallucis brevis.

1) Extensor digitorum brevis muscle:

- Origin:
 - the superior surface of the calcaneal process.
 - The extensor retinaculum.
- Body: short, thin, flattened, fan-shaped muscle.
- Insertion: by three tendons that join the lateral borders of the second, third, and fourth tendons of the extensor digitorum longus.
- Action: Extensor of the 2nd, 3rd, and 4th toes.
- Innervation: Deep fibular nerve (sacral plexus).

2) Extensor hallucis brevis muscle:

- Origin:
 - The superior surface of the calcaneal process.
 - The extensor retinaculum.

- Body : Short and flat muscle.
- Insertion: The base of the proximal phalanx (P1) of the hallux.
- Action: Extensor of the hallux.
- Innervation: Deep fibular nerve (sacral plexus).

B. MUSCLES OF THE PLANTAR REGION:

Numbering 19 muscles, divided into 3 groups:

1. **Middle (intermediate) plantar group:** 13 muscles arranged in 3 layers: deep, middle, and superficial.

❖ **Deep layer:** interosseous compartment: 7 muscles

- 4 dorsal interosseous muscles
- 3 plantar interosseous muscles

a) Dorsal interosseous muscles:

- Origin: the lateral and medial surfaces of the two metatarsals that define the interosseous space.
- Body : short, prismatic, and triangular muscles.
- Insertion:
 - the 1st and 2nd interosseous muscles on the proximal phalanx (P1) of the 2nd toe.
 - the 3rd interosseous muscle on the proximal phalanx (P1) of the 3rd toe.
 - the 4th interosseous muscle on the proximal phalanx (P1) of the 4th toe.
- Action: Flexes the proximal phalanges (P1) of the toes and abducts the toes.
- Innervation: Lateral plantar nerve (sacral plexus).

b) Plantar interosseous muscles:

- Origin: Medial surfaces of the last three metatarsals.
- Body: Short, prismatic muscles.
- Insertion: Base of the proximal phalanx (P1) of the corresponding toe.
- Action: Flexes the proximal phalanges (P1) of the toes and adducts the toes.
- Innervation: Lateral plantar nerve (sacral plexus).

❖ **Middle layer:** 5 muscles associated with the flexor digitorum longus.

- 4 lumbrical muscles.
- The quadratus plantae muscle.

a) Lumbrical muscles:

- Origin: The tendons of the flexor digitorum longus; each muscle inserts onto two tendons of the flexor digitorum longus.
- Body: Short, fusiform muscles.
- Insertion: The base of the proximal phalanx (P1), an expansion onto the tendon of the extensor digitorum longus.
- Action: Flexes P1 and extends P2 and P3.
- Innervation: Medial and lateral plantar nerves (sacral plexus).

b) Quadratus plantae muscle:

- Origin: Two heads:
 - Medial head: On the calcaneal groove and the posteromedial tuberosity of the calcaneus.
 - Lateral head: On the posterolateral tuberosity of the calcaneus.
- Body: Short, flat, triangular muscle.
- Insertion: On the tendon of the flexor digitorum longus.
- Action: Corrects toe obliquity caused by the flexor digitorum longus.
- Innervation: Medial and lateral plantar nerves (sacral plexus).

❖ **Superficial plane:** 01 single muscle:

a) **Flexor digitorum brevis muscle:**

- Origin: the posteromedial tuberosity of the inferior surface of the calcaneus, the medial and lateral plantar intermuscular septa.
- Body: long, thin, and flat muscle.
- Insertion: by four tendinous tendons inserting onto the second phalanx (P2) of the four ulnar toes.
- Action: flexes the P2 of the four ulnar toes on the proximal phalanx (P1), then the proximal phalanx on the metatarsal bone.
- Innervation: medial plantar nerve (sacral plexus).

2. **MEDIAL PLANTAR GROUP:**

a) **Flexor hallucis brevis muscle:**

- Origin: on the lateral and intermediate cuneiform bones, and on the cuboid.
- Body: short muscle, thick posteriorly and bifurcated anteriorly.
- Insertion:
 - medial tendon on the medial sesamoid bone and the base of the proximal phalanx (P1).
 - Lateral tendon: on the lateral sesamoid bone and the base of the proximal phalanx (P1).
- Action: flexes the hallux.
- Innervation: medial plantar nerve (sacral plexus).

b) **Adductor hallucis muscle:**

Located lateral to the flexor hallucis brevis, it is composed of two heads: oblique and transverse.

- Origin:
 - Oblique head: on the cuboid bone, the third cuneiform bone, and the base of the third and fourth metatarsal bones.
 - Transverse head: on the glenoid ligament of the third, fourth, and fifth metatarsophalangeal joints.
- Insertion: on the lateral sesamoid bone and the base of the proximal phalanx (P1) of the hallux.
- Action: flexes and adds to the hallux.
- Innervation: lateral plantar nerve (sacral plexus).

c) **Abductor hallucis muscle:**

- Origin: on the posteromedial tuberosity of the inferior surface of the calcaneus.
- Body: elongated and flattened muscle.
- Insertion: on the medial sesamoid bone and the base of the proximal phalanx (P1) of the hallux.
- Action: flexes and abducts the hallux.
- Innervation: medial plantar nerve (sacral plexus).

3. **LATERAL PLANTAR GROUP:**

a) **Flexor digiti minimi brevis muscle:**

- Origin: on the inferior surface of the cuboid bone and the base of the 5th metatarsal.
- Body: fusiform muscle.
- Insertion: on the base of the proximal phalanx (P1) of the 5th toe.
- Action: flexes the proximal phalanx (P1) of the 5th toe.
- Innervation: lateral plantar nerve (sacral plexus).

b) **Opposition digiti minimi muscle:**

- Origin: the inferior surface of the cuboid bone.
- Belly: short and flat muscle.
- Insertion: the lateral border of the 5th metatarsal.
- Action: deflects the 5th metatarsal medially.
- Innervation: lateral plantar nerve (sacral plexus).

c) **Abductor pollicis muscle:**

- Origin: on the posterolateral tuberosity of the inferior surface of the calcaneus and the lateral tuberosity of the 5th metatarsal.
- Belly: elongated muscle, thick and tendinous anteriorly.
- Insertion: on the base of the proximal phalanx (P1) of the little toe.
- Action: flexes and abducts the little toe.
- Innervation: lateral plantar nerve (sacral plexus).

Reference :

- Rouviere H. Anatomie Humaine Descriptive, Topographique et Fonctionnelle Tome3.Membres ed. Masson 2002. (traduit en anglais).
- Hammoudi SS. Le cours d'anatomie. Fasc.2 appareil locomoteur. Membre inférieur. ISBN 2008(traduit en anglais).
- Kamina .Myologie du membre inférieur. 2ème édition. Maloine. (traduit en anglais).



